

LÉONARD Cédric

08/03/1999

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cedric.leonard@dlr.de

Ismaningerstr.,
81675, Munich

Google Scholar

MISCELLANEOUS SKILLS

- Programming (Github):
- Teamwork and communication
- Scientific literature review
- Popularization and teaching

LANGUAGES

- French: Native
- English: C2
- German: B2

HOBBIES & INTERESTS

- All kinds of mountain activities
- 5 years of theater
- 13 years of judo - black belt
- Oversharing my love for science

EDUCATION

- **Since 2022:** PhD at TUM and DLR, Munich, Germany
- **2021 / 2022**: Double Degree, Master of Science (Technology) Åbo Akademi University, Finland
- **2017 / 2021**: Electronics and Industrial Computer Engineering, INSA de Rennes, France
- **2013 / 2017**: High school diploma, focus on engineering sciences

PROFESSIONAL EXPERIENCES

Since October 2022, Munich // Deutsches Zentrum für Luft- und Raumfahrt (DLR)

- PhD at the Remote Sensing Technology Institute (DLR)
  - *"Learned SAR Despeckling and Compression on Embedded FPGA"*
  - Learned Image Compression (LIC) + TerraSAR-X despeckling
  - FPGA SoC design using AMD's DPU

January – Mars 2026, Frascati // European Space Agency (ESA) - ESRIN

- Visiting Researcher at ESA Φ-lab
  - *"SAR RCMC data compression using hyperprior autoencoders."*
  - Research sharing in multiple research groups
  - Networking and collaboration

June – August 2021, Rennes // IETR

- Research Internship in the VAADER group
  - "*Tangled Program Graphs for Low Complexity Block Partitioning in Video Coding*"
  - Genetic Programming (**TPGs**) to reduce the computational complexity of a video encoding standard (**VVC**);

June - July 2020, Nantes // Naostage

- Intern at Naostage
  - Handling and development of **Qaterial** (C++ and Qt graphic library)
  - Conception of PsnViewer: real-time monitoring of **PSN** packets.

TEACHING AND ASSOCIATIVE EXPERIENCES

Since 2023, Munich // TUM, CAPS chair

- Teaching and supervision
  - Bachelor's and Master's theses supervision
  - Seminars organization

2017 - 2021, INSA de Rennes // Amicale de l'INSA de Rennes

- Member of the Sound&Light club (Presidency 2019 - 2020)
  - Setup and technical support on every internal event at the university (festivals, concerts, shows, conferences, etc.);
  - Design and implementation of lighting effects with **GrandMa2.**

2016 and 2019-2020

- Tutoring
  - Tutor for a support group of 2nd year students;
  - Mathematics and physics courses for high school students.



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SCIENTIFIC TALKS & PUBLISHED CONTRIBUTIONS

June 2026, Baden-Baden // EUSAR 2026

- Conference talk and paper
  - *"Towards end-to-end SAR raw data handling: RCMC data compression using hyperprior autoencoders"*
  - Learned Image Compression for partially focused SAR data
  - IEEE Xplore

April 2026 // ACM Computing Surveys

- Journal paper
  - *"FPGA-Enabled Machine Learning Applications in Earth Observation: A Systematic Review"*
  - Systematic survey paper over RS, ML, and FPGA research
  - ACM CSUR

April 2025, Vienna & Innsbruck // LPS 2025

- Conference talk
  - *"Onboard Machine Learning-based Compression of Synthetic Aperture Radar (SAR) Images Using FPGA/MPSoC Hardware"*
  - Deploying a joint denoising and compression method on the edge
  - DLR elib

April & October 2024, Vienna & Innsbruck // EGU & MARS 2024

- Poster (Natural Hazards) & Oral presentation
  - *"SAR GRD Data Compression using Mean-Scale Hyperprior architectures"*
  - Benchmarking autoencoder-based architectures on simplistic EO datasets: EuroSAT and EuroSAT-SAR
  - EGU 2024

January – June 2022, Turku // Åbo Akademi

- Master's Thesis in Computer Engineering
  - "*Performance exploration of the REST API using online GAN*"
  - Literature review about Deep Neural Networks, Generative Adversarial Networks, **REST APIs,** and Performance Testing
  - Exploring the possibilities of automated test generation of a new Online learning algorithm based on **GANs**
  - Åbo Akademi doria

AWARDS AND GRANTS

January – March 2026, Research Stay at ESA, Frascati, Italy

- DAAD Forschungsstipendien für Doktorandinnen und Doktoranden, 2025
  - *~5.000€ for the duration of the stay.*